

Memo Report

October 2, 2025

TO: Erin Engineer, P.E., Director, Abiquiu Reservoir Hydropower Feasibility Study Team

FROM: Sachi Thakur @ Illinois, Student Intern

RE: Potential impact of seasonal flow variation on hydropower potential

CC: CEE 190 Instructional Team

This analysis aimed to assess seasonal trends in the potential power output of the proposed Abiquiu Reservoir hydropower facility. The data that was compared was the potential power of the hydroelectric generator in kilowatt hours over the years 2019 to 2024. The data was collected every month, allowing for a clear distinction between the performance as seasons alternate. This data of the power potential is reported by the U.S. Geological Survey gaging station. The reports and patterns from the graph over 5 years show that the hydroelectric generation was at peak during the late winters and early spring months. The lowest generation is during the hot summer months due to the lack of water. The generation peaks during these times due to the melting snow and rain during the months of February through April.

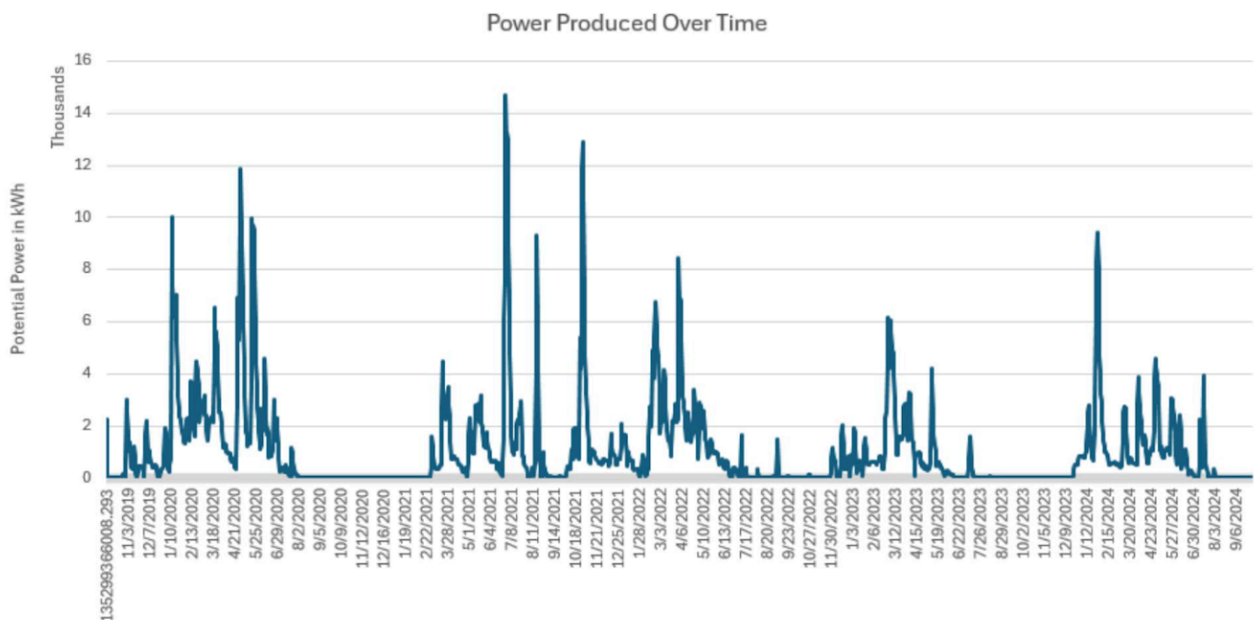


Figure 1. Line graph showing relationship between potential power of hydroelectric generator and seasonal months